

Khalid Rabbani

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EDUCATION

BMS College of Engineering

B.E in Computer Science & Engineering

Bangalore, India

2023 – 2027

Noble International School

Class 12 (CBSE)

Doha, Qatar

2023

ABOUT ME

3rd-year Computer Science student with experience in Data Analytics, Data Science and Machine Learning. Skilled in Python (Pandas, NumPy), SQL, Excel, Power BI, Machine Learning and data visualization tools, HTML, CSS, JS.I focus on analyzing data, extracting meaningful patterns, and applying machine learning techniques to solve practical problems.

PROJECTS

Cancer Tumor Diagnostic

Technologies: Python, Scikit-learn, XGBoost, Pandas, NumPy

- Built a classification pipeline comparing 5 ML models (Logistic Regression, KNN, SVM, Random Forest, XGBoost) to predict breast tumor malignancy from 30 cell nucleus features
- Achieved 97.37% accuracy and 92.86% recall using SVM; prioritized recall as the key metric given the high cost of false negatives in medical diagnosis.
- Optimized SVM using GridSearchCV with 5-fold cross-validation; identified area_worst and concave_points_worst as top predictors using Random Forest feature importance.

Sports Injury & Performance Impact Analysis

Technologies: Python, Excel, Power BI

- Analyzed 600+ Premier League injury records (2019–2023) to assess the impact of injuries on player performance and team outcomes.
- Performed data cleaning and feature engineering using Python(Pandas, NumPy), including handling missing values and deriving pre/post-injury performance metrics.
- Built an interactive Power BI dashboard to analyze injury trends, recovery duration, and performance impact across teams and player demographics.

Blink2Speech

Technologies: Python, OpenCV, MediaPipe, Tkinter, Text-to-Speech (pyttsx3)

- Built a real-time assistive tool that detects eye blinks using MediaPipe's ML-powered facial landmark tracking and translates them into Morse code for users with paralysis or motor impairments.
- Implemented calibrated blink classification (short vs long), Morse decoding with letter/word timing, and emergency quick-command support.
- Developed a full Tkinter GUI with live camera feed, blink metrics, real-time decoding preview, and text-to-speech output.

SKILLS

Technical Skills: Python (Pandas, NumPy), SQL, Excel, HTML, CSS, JS

Data Science: Machine Learning (Scikit-learn), Data Cleaning, Exploratory Data Analysis, Data Visualization

Tools: PowerBI, Git, GitHub, VS Code

Soft Skills: Communication, Teamwork, Leadership, Problem-Solving, Adaptability

LEADERSHIP & ACTIVITIES

Senior Coordinator, CodeIO Club (BMSCE) — Organizing technical events, mentoring juniors, and coordinating club activities.